《计算机网络技术及应用》
实验四报告

申报学校：福建理工大学

学　　院：计算机科学与数学学院

课程类型：专业基础课

适用专业：网络空间安全22

授课教师：嵇晓蓉

实验四 VLSM 复杂静态路由配置

一、实验目的

1) 能解释 VRP 的基本操作命令的含义，并熟练运用

2) 能进行 VLSM 的合理的子网划分

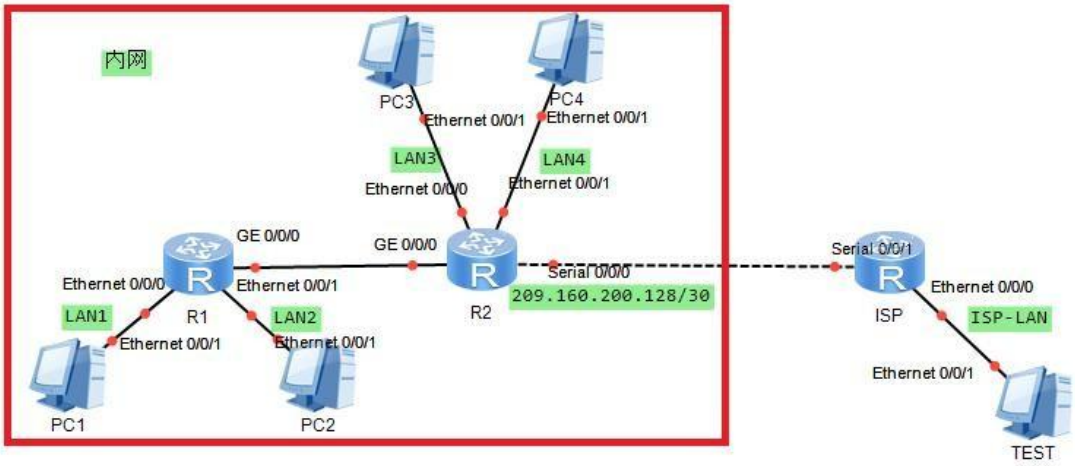
3) 能分析总结路由并阐释其作用

4) 能分析 PC 机之间是如何通过路由器进行通信的

5) 能分析内外网之间的通信过程

二、实验内容

1、场景



场景说明：R1 和 R2 是内网路由器，通过 ISP 连接到 Internet，TEST 为因特网上的测试主机，该主机地址可以采用任意的外网地址(本场景忽略 NAT)。

二、实验要求

1) 使用 10.20.30.0/24 网络划分子网，LAN1 需要 30 个主机，LAN2 需要 12 个主机，LAN3 和 LAN4 各需要 50 个主机，为内网各个子网划分网络号，ISP-LAN 的地址可以自己定义任意外网地址。

2) 每个子网的第 1 个有效地址分配给网关，最后一个有效地址分配给主机

3) 路由配置要求：在 R1,R2 和 ISP 上配置静态路由，注意 ISP 上配置总结路由，R1 和 R2 上需要配置默认路由访问外网

4) 连通要求：内外网所有的主机之间都可以访问

5) 路由器改名要求：” 学号-R1” ,” 学号-R2” ,” 学号-ISP”

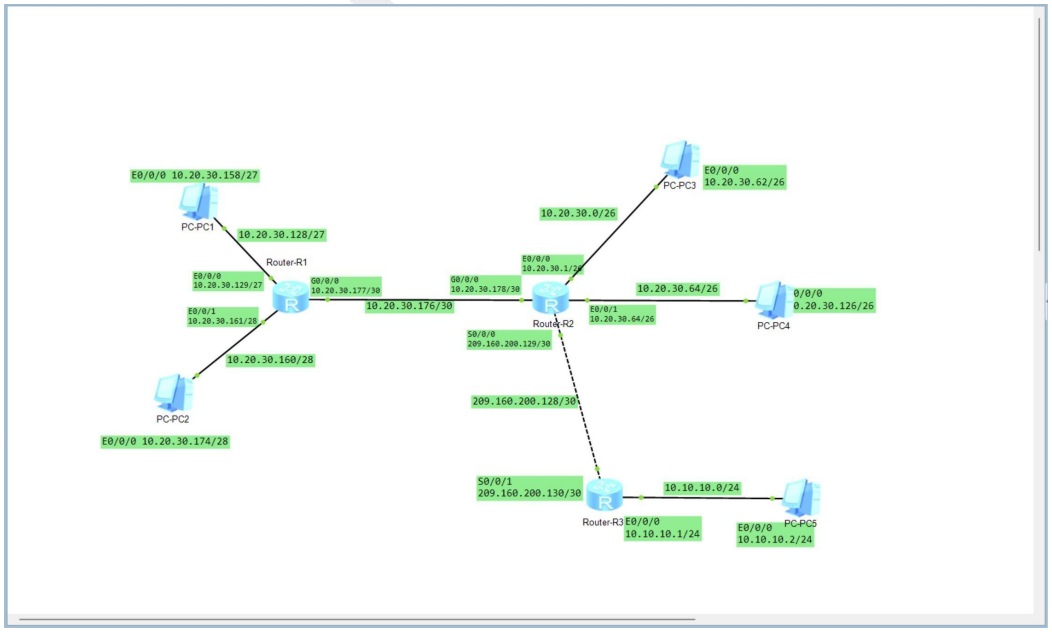
三、实验记录

1、根据子网划分的要求，填写下列地址表

3-1 IP 地址表				
设备	接口	IP 地址	子网掩码	默认网关
R1	E0/0/0	10.20.30.129	255.255.255.224	无
	E0/0/1	10.20.30.161	255.255.255.240	无
	G0/0/0	10.20.30.177	255.255.255.252	无
R2	E0/0/0	10.20.30.1	255.255.255.192	无
	E0/0/1	10.20.30.65	255.255.255.192	无
	G0/0/0	10.20.30.178	255.255.255.252	无
	S0/0/0	209.160.200.129	255.255.255.252	无
ISP	E0/0/0	10.10.10.1	255.255.255.0	无
	S0/0/1	209.160.200.130	255.255.255.252	无
PC1	网卡	10.20.30.158	255.255.255.224	10.20.30.129
PC2	网卡	10.20.30.174	255.255.255.240	10.20.30.161
PC3	网卡	10.20.30.62	255.255.255.192	10.20.30.1
PC4	网卡	10.20.30.126	255.255.255.192	10.20.30.65
TEST	网卡	10.10.10.2	255.255.255.0	10.10.10.1

2、贴出每个路由器的配置,包括地址和路由（模式和有效信息）

拓扑：



R1:

```
<Huawei>sy
[Huawei]sysname 3220812128-R1
[3220812128-R1]int g0/0/0
[3220812128-R1-GigabitEthernet0/0/0]ip add 10.20.30.177 30
[3220812128-R1-GigabitEthernet0/0/0]int e0/0/0
[3220812128-R1-Ethernet0/0/0]ip add 10.20.30.129 27
[3220812128-R1-Ethernet0/0/0]int e0/0/1
[3220812128-R1-Ethernet0/0/1]ip add 10.20.30.161 28
[3220812128-R1-Ethernet0/0/1]q
[3220812128-R1]ip route-static 0.0.0.0 0 10.20.30.178
```

R2:

```
<Huawei>sy
[Huawei]sysname 3220812128-R2
[3220812128-R2]int g0/0/0
[3220812128-R2-GigabitEthernet0/0/0]ip add 10.20.30.178 30
[3220812128-R2-GigabitEthernet0/0/0]int e0/0/0
[3220812128-R2-Ethernet0/0/0]ip add 10.20.30.1 26
[3220812128-R2-Ethernet0/0/0]int e0/0/1
[3220812128-R2-Ethernet0/0/1]ip add 10.20.30.65 26
[3220812128-R2-Ethernet0/0/1]int s0/0/0
[3220812128-R2-Serial0/0/0]ip add 209.160.200.129 30
[3220812128-R2-Serial0/0/0]q
[3220812128-R2]ip route-static 0.0.0.0 0 209.160.200.130
[3220812128-R2]ip route-static 10.20.30.128 27 10.20.30.177
[3220812128-R2]ip route-static 10.20.30.160 28 10.20.30.177
```

ISP:

```
<Huawei>sy
[Huawei]sysname 3220812128-ISP
[3220812128-ISP]int e0/0/0
[3220812128-ISP-Ethernet0/0/0]ip add 10.10.10.1 24
[3220812128-ISP-Ethernet0/0/0]int s0/0/1
[3220812128-ISP-Serial0/0/1]ip add 209.160.200.130 30
[3220812128-ISP-Serial0/0/1]q
[3220812128-ISP]ip route-static 10.20.30.0 24 209.160.200.129
```

3、贴出路由器和主机的 IP 地址的检测结果（模式和有效信息）

R1:

```
[3220812128-R1]dis ip int brief
*down: administratively down
!down: FIB overload down
^down: standby
(l): loopback
(s): spoofing
(d): Dampening Suppressed
The number of interface that is UP in Physical is 4
The number of interface that is DOWN in Physical is 7
The number of interface that is UP in Protocol is 4
The number of interface that is DOWN in Protocol is 7

Interface                         IP Address/Mask      Physical   Protocol
Ethernet0/0/0                     10.20.30.129/27      up         up
Ethernet0/0/1                     10.20.30.161/28      up         up
GigabitEthernet0/0/0              10.20.30.177/30      up         up
GigabitEthernet0/0/1              unassigned           down       down
GigabitEthernet0/0/2              unassigned           down       down
GigabitEthernet0/0/3              unassigned           down       down
NULL0                             unassigned           up         up(s)
Serial0/0/0                       unassigned           down       down
Serial0/0/1                       unassigned           down       down
Serial0/0/2                       unassigned           down       down
Serial0/0/3                       unassigned           down       down
```

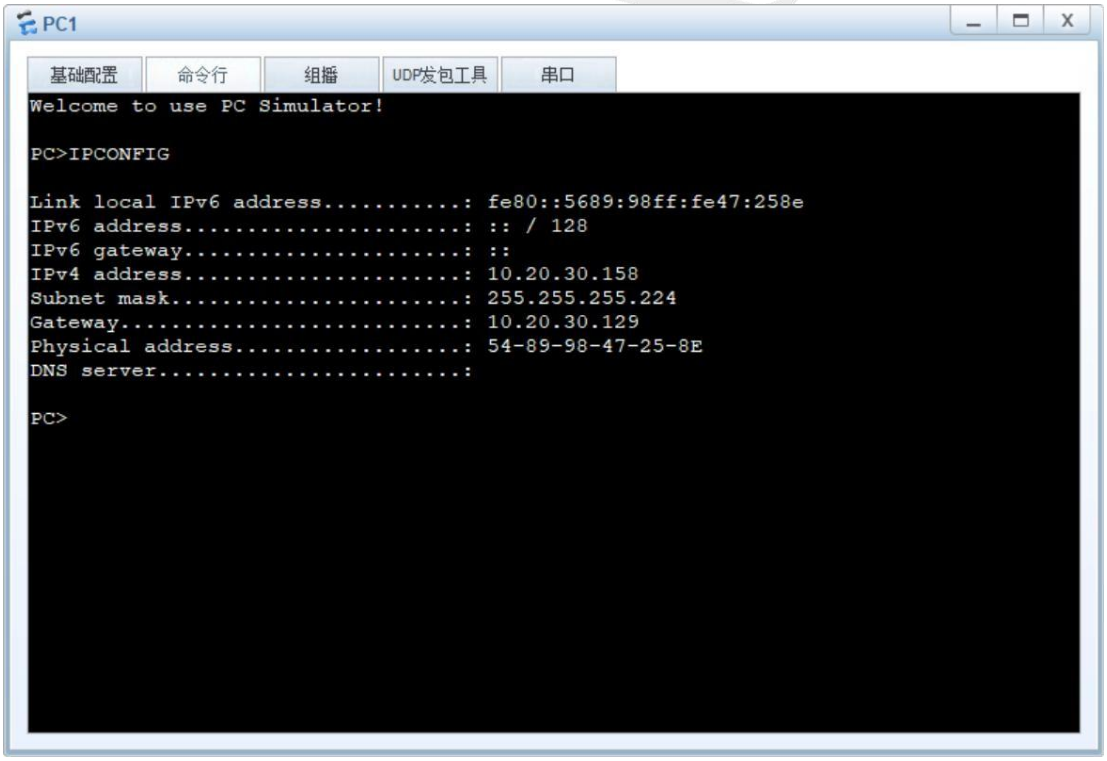
R2:

```
[3220812128-R2]dis ip int b
*down: administratively down
!down: FIB overload down
^down: standby
(l): loopback
(s): spoofing
(d): Dampening Suppressed
The number of interface that is UP in Physical is 5
The number of interface that is DOWN in Physical is 6
The number of interface that is UP in Protocol is 5
The number of interface that is DOWN in Protocol is 6

Interface                         IP Address/Mask      Physical   Protocol
Ethernet0/0/0                     10.20.30.1/26        up         up
Ethernet0/0/1                     10.20.30.65/26       up         up
GigabitEthernet0/0/0              10.20.30.178/30      up         up
GigabitEthernet0/0/1              unassigned           down       down
GigabitEthernet0/0/2              unassigned           down       down
GigabitEthernet0/0/3              unassigned           down       down
NULL0                             unassigned           up         up(s)
Serial0/0/0                       209.160.200.129/30   up         up
Serial0/0/1                       unassigned           down       down
Serial0/0/2                       unassigned           down       down
Serial0/0/3                       unassigned           down       down
```

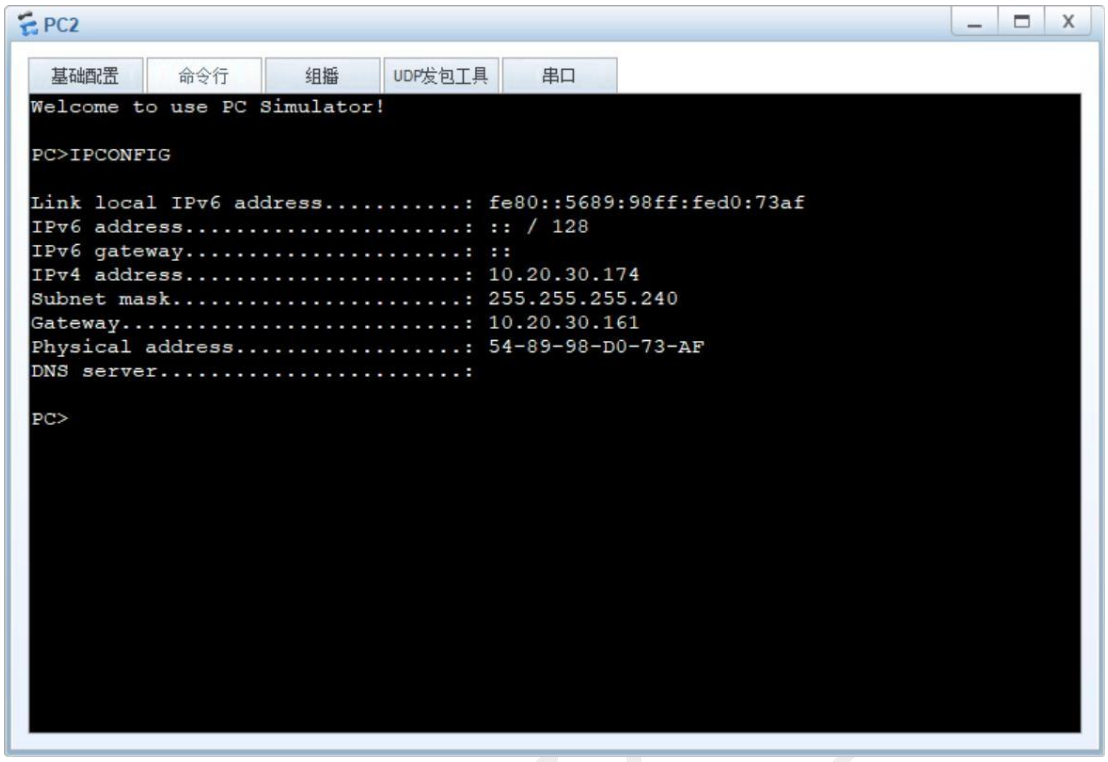
ISP:

```
[3220812128-ISP]dis ip int b
*down: administratively down
!down: FIB overload down
^down: standby
(l): loopback
(s): spoofing
(d): Dampening Suppressed
The number of interface that is UP in Physical is 3
The number of interface that is DOWN in Physical is 8
The number of interface that is UP in Protocol is 3
The number of interface that is DOWN in Protocol is 8

Interface                         IP Address/Mask      Physical   Protocol
Ethernet0/0/0                     10.10.10.1/24        up         up
Ethernet0/0/1                     unassigned           down       down
GigabitEthernet0/0/0              unassigned           down       down
GigabitEthernet0/0/1              unassigned           down       down
GigabitEthernet0/0/2              unassigned           down       down
GigabitEthernet0/0/3              unassigned           down       down
NULL0                             unassigned           up         up(s)
Serial0/0/0                       unassigned           down       down
Serial0/0/1                       209.160.200.130/30   up         up
Serial0/0/2                       unassigned           down       down
Serial0/0/3                       unassigned           down       down
```

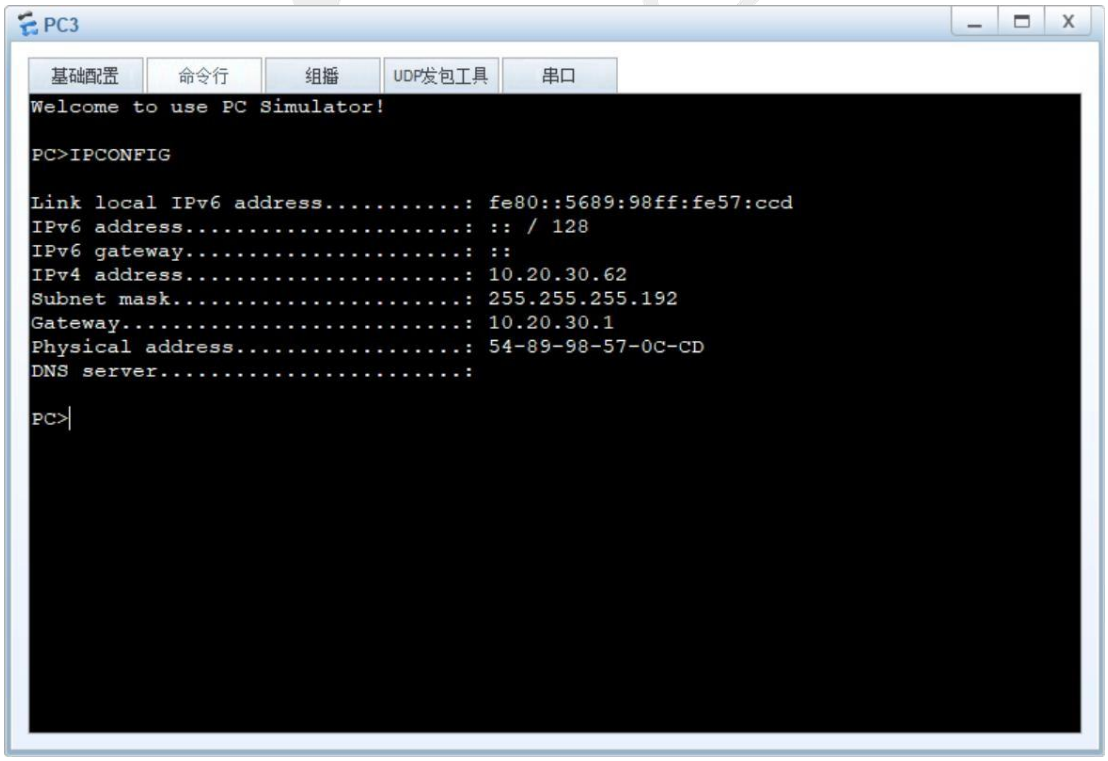
PC1:



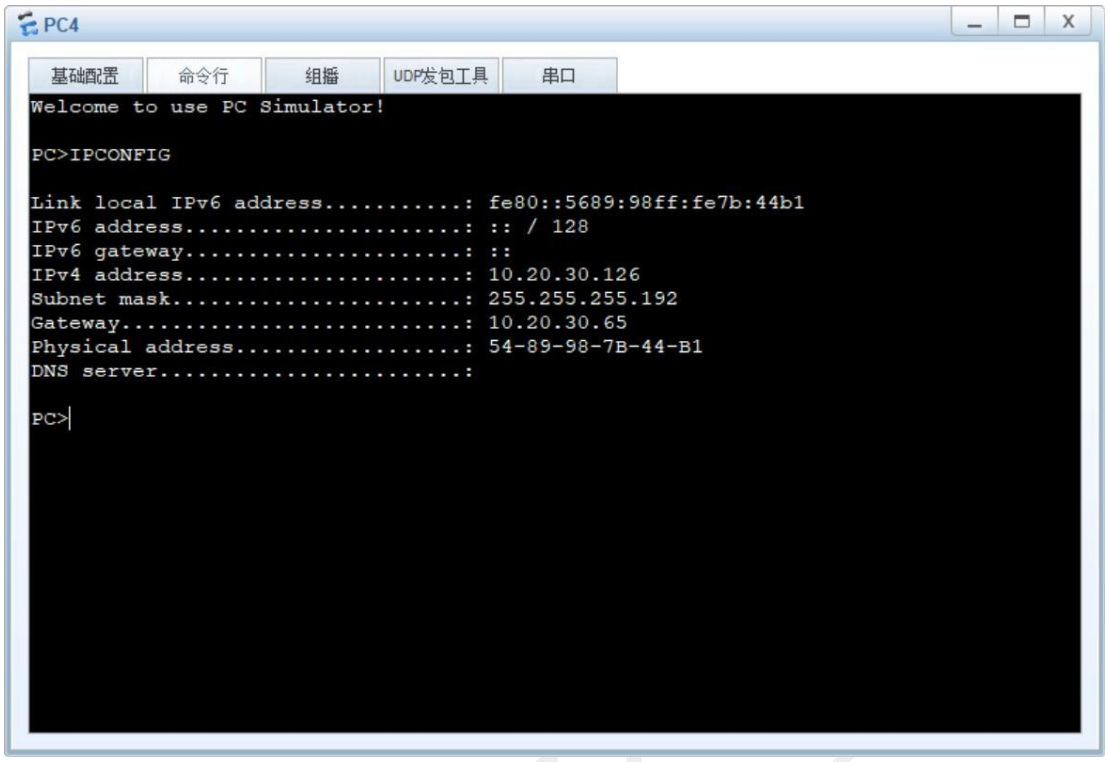
PC2:


PC2
基础配置
命令行
组播
UDP发包工具
串口
Welcome to use PC Simulator!
PC>IPCONFIG
Link local IPv6 address.....: fe80::5689:98ff:fed0:73af
IPv6 address.....: :: / 128
IPv6 gateway.....: ::
IPv4 address.....: 10.20.30.174
Subnet mask.....: 255.255.255.240
Gateway.....: 10.20.30.161
Physical address.....: 54-89-98-D0-73-AF
DNS server.....:
PC>

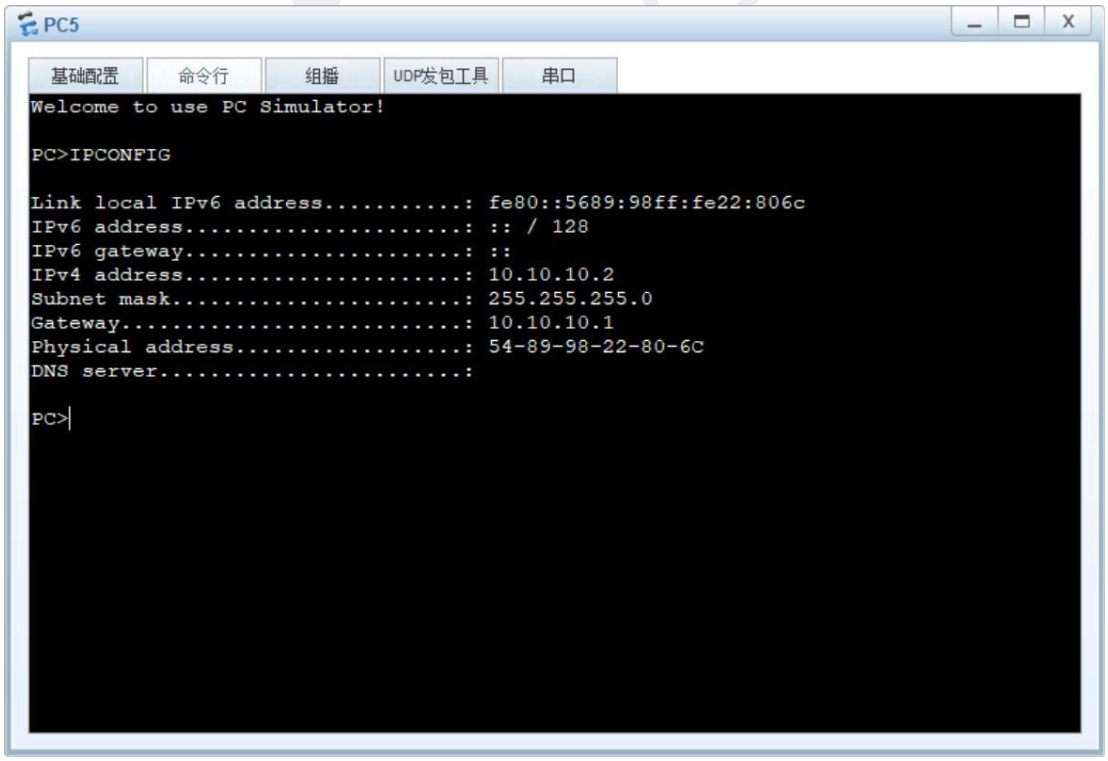
PC3:


PC3
基础配置
命令行
组播
UDP发包工具
串口
Welcome to use PC Simulator!
PC>IPCONFIG
Link local IPv6 address.....: fe80::5689:98ff:fe57:ccd
IPv6 address.....: :: / 128
IPv6 gateway.....: ::
IPv4 address.....: 10.20.30.62
Subnet mask.....: 255.255.255.192
Gateway.....: 10.20.30.1
Physical address.....: 54-89-98-57-0C-CD
DNS server.....:
PC>

PC4:



PC5:



4、贴出路由器路由表并分析（模式和有效信息）

R1:

```
[3220812128-R1]DIS IP routing-table
Route Flags: R - relay, D - download to fib
------------------------------------------------------------------------------
Routing Tables: Public
         Destinations : 9        Routes : 9

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface

        0.0.0.0/0   Static  60   0          RD   10.20.30.178    GigabitEthernet
0/0/0
   10.20.30.128/27  Direct  0    0           D   10.20.30.129    Ethernet0/0/0
   10.20.30.129/32  Direct  0    0           D   127.0.0.1       Ethernet0/0/0
   10.20.30.160/28  Direct  0    0           D   10.20.30.161    Ethernet0/0/1
   10.20.30.161/32  Direct  0    0           D   127.0.0.1       Ethernet0/0/1
   10.20.30.176/30  Direct  0    0           D   10.20.30.177    GigabitEthernet
0/0/0
   10.20.30.177/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/0
      127.0.0.0/8   Direct  0    0           D   127.0.0.1       InLoopBack0
      127.0.0.1/32  Direct  0    0           D   127.0.0.1       InLoopBack0
```

在 R1 中除了路由器自己学习的直连路由，还有我们手动配置的默认路由0.0.0.0，通过这个默认路由，LAN1 和 LAN2的主机都能将数据交给R2转发，同时也能访问R2下的子网LAN3、LAN4。

R2:

```
[3220812128-R2]DIS IP routing-table
Route Flags: R - relay, D - download to fib
------------------------------------------------------------------------------
Routing Tables: Public
         Destinations : 14       Routes : 14

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface

        0.0.0.0/0   Static  60   0          RD   209.160.200.130 Serial0/0/0
     10.20.30.0/26  Direct  0    0           D   10.20.30.1      Ethernet0/0/0
     10.20.30.1/32  Direct  0    0           D   127.0.0.1       Ethernet0/0/0
    10.20.30.64/26  Direct  0    0           D   10.20.30.65     Ethernet0/0/1
    10.20.30.65/32  Direct  0    0           D   127.0.0.1       Ethernet0/0/1
   10.20.30.128/27  Static  60   0          RD   10.20.30.177    GigabitEthernet
0/0/0
   10.20.30.160/28  Static  60   0          RD   10.20.30.177    GigabitEthernet
0/0/0
   10.20.30.176/30  Direct  0    0           D   10.20.30.178    GigabitEthernet
0/0/0
   10.20.30.178/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/0
      127.0.0.0/8   Direct  0    0           D   127.0.0.1       InLoopBack0
      127.0.0.1/32  Direct  0    0           D   127.0.0.1       InLoopBack0
209.160.200.128/30  Direct  0    0           D   209.160.200.129 Serial0/0/0
209.160.200.129/32  Direct  0    0           D   127.0.0.1       Serial0/0/0
209.160.200.130/32  Direct  0    0           D   209.160.200.130 Serial0/0/0
```

在 R2 中，我们配置了一条通过 s0/0/0 访问外网的默认路由0.0.0.0，这样使得LAN3、LAN4和通过R2转发的主机都能访问外网，同时添加了 LAN1 和 LAN2 的静态路由10.20.30.128/27、10.20.30.160/28通过10.20.30.177转发，让内网主机都能互相访问。

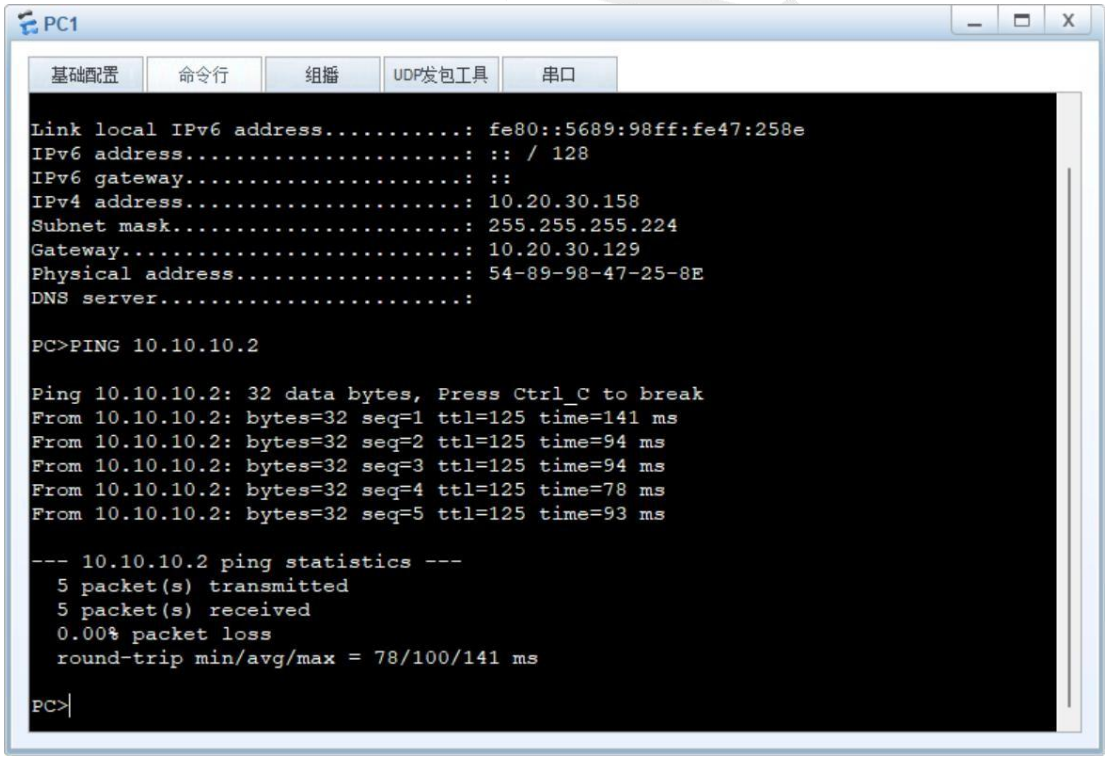
ISP:

```
[3220812128-ISP]DIS IP routing-table
Route Flags: R - relay, D - download to fib
------------------------------------------------------------------------------
Routing Tables: Public
         Destinations : 8        Routes : 8

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface

     10.10.10.0/24  Direct  0    0           D   10.10.10.1      Ethernet0/0/0
     10.10.10.1/32  Direct  0    0           D   127.0.0.1       Ethernet0/0/0
     10.20.30.0/24  Static  60   0          RD   209.160.200.129 Serial0/0/1
      127.0.0.0/8   Direct  0    0           D   127.0.0.1       InLoopBack0
      127.0.0.1/32  Direct  0    0           D   127.0.0.1       InLoopBack0
209.160.200.128/30  Direct  0    0           D   209.160.200.130 Serial0/0/1
209.160.200.129/32  Direct  0    0           D   209.160.200.129 Serial0/0/1
209.160.200.130/32  Direct  0    0           D   127.0.0.1       Serial0/0/1
```

在 ISP 中，我们加了一条汇总路由，只要是访问 10.20.30.0/24 的网段的都从 R2 的 s0/0/0 进入，实现了外网的主机也能访问内网主机的功能

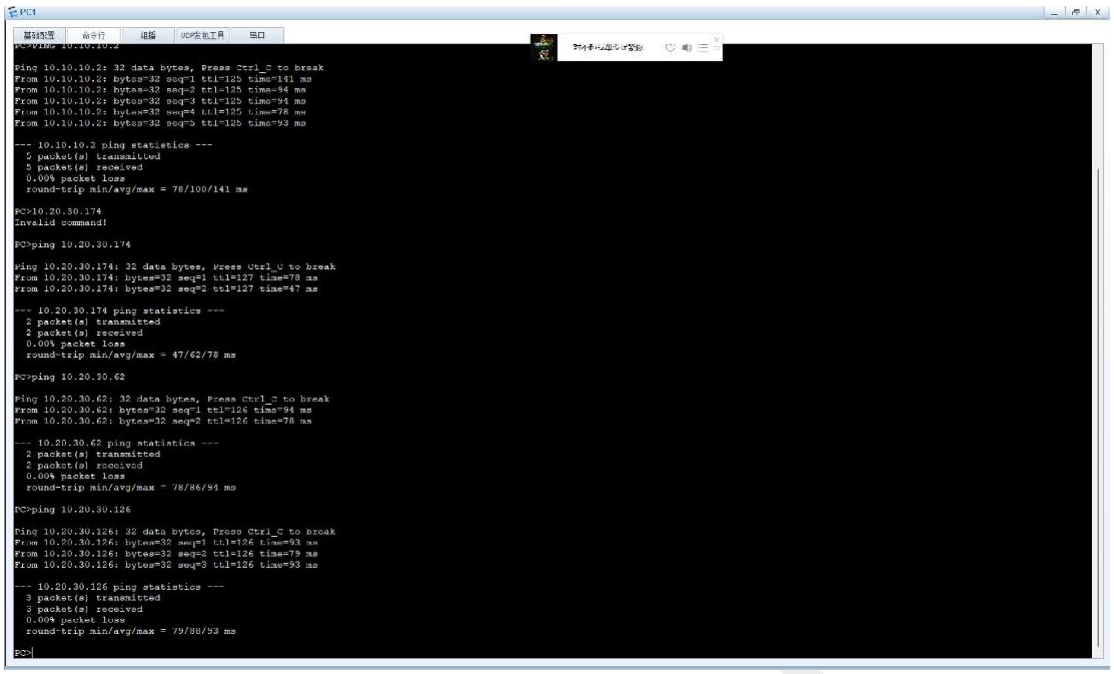
5、联通性测试结果，并给出结论

用内网主机 pc1ping 外网主机 pc5：



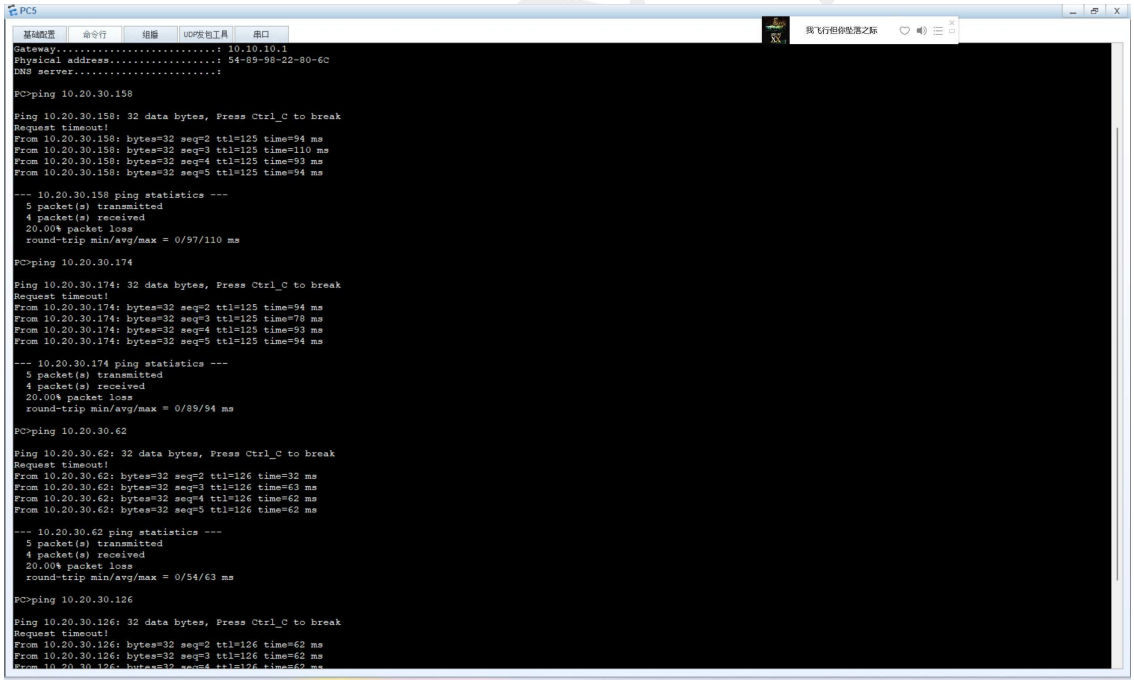
成功连通

内网主机互相访问（用 pc1ping 内网其他主机）：



均可连通

外网主机 pc5 访问内网主机：



均可连通

6、排错总结

基本上没有问题，就是要在子网划分的时候要小心，有的时候会搞乱网段。通过本次实验，我能更加熟悉静态的路由的配置方法，以及如何正确合理的区划分子网，并且了解了如何去实现内网和外网的通信。同时也让我对 ensp 模拟器的命令更加的熟悉，对于路由器和主机的配置更加熟练。

jixr